

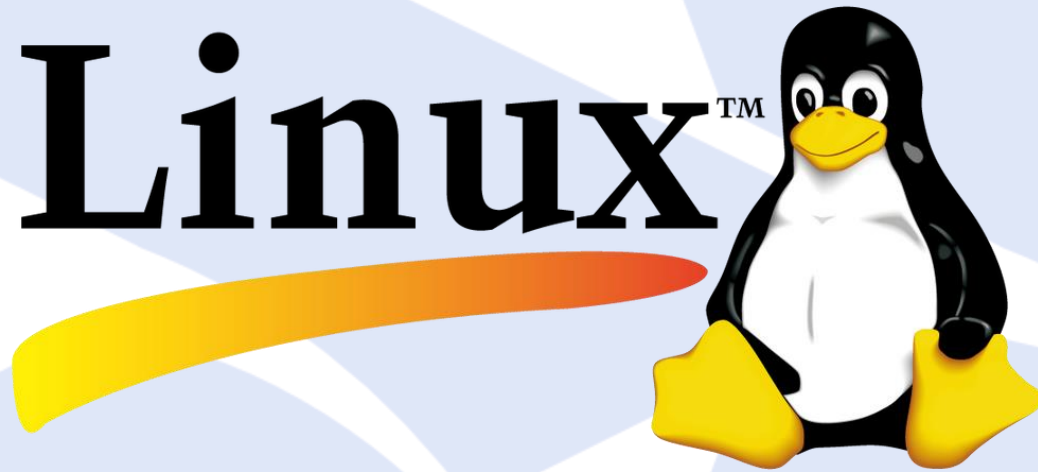
Route

Linux(RHEL) - Introduction

2025



LinuxTM



Session One

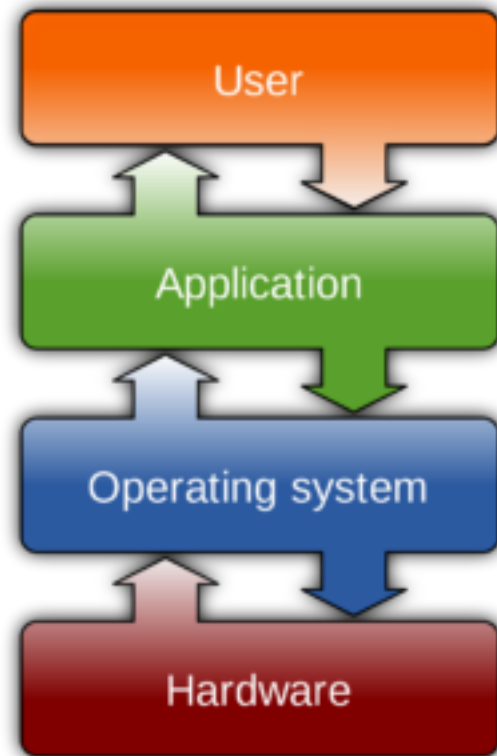
Getting Started

- 1 Operating System
- 2 Linux
- 3 Open Source
- 4 Distributions

Operating System

Definition

AN Operating systems (OS) is a software acts as an interface between the hardware component and the user. Every computer must have at least one operating system to run other programs. Applications like (Browser , MS Office , notepad, Games,...etc) needs an environment to run and perform its tasks



Objectives of OS

- 1 Convenience
- 2 Efficiency
- 3 Ability to Evolve
- 4 Management of system resources

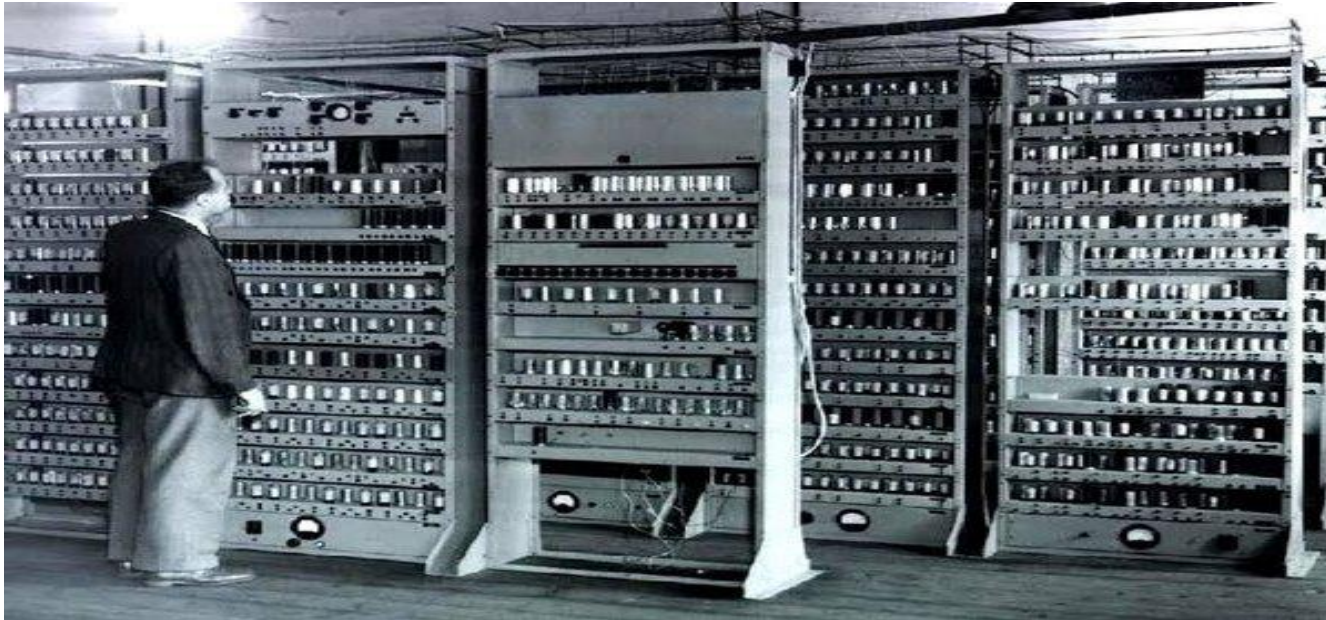
Functionalities of OS

- 1 Memory Management
- 2 Process Management/Scheduling
- 3 File System Management
- 4 Device Management
- 5 Security/Privacy Management

History

1. First Generation (1940– 1956) – vacuum tubes

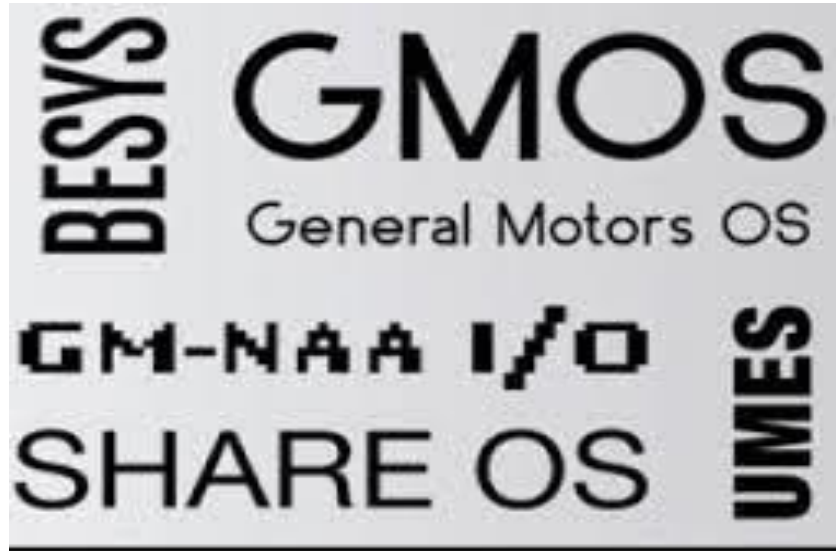
In the first electronic computer was developed in 1940, it was created without any operating system



History

2. Second Generation (1955– 1965) – General Motors OS (GMOS)

In the first Operating system (OS) was created at the early of 1950s and was known as GMOS. General Motors has developed OS for the IBM Computers.



History

3. Third Generation (1965– 1980)

- IBM developed for 360/ systems Mainframes.
- Unix (1971) was developed by AT&T Bell laboratories
- Unix revolutionized OS design with its simplicity, portability, and multitasking capabilities.
- Personal computers emerged, leading to simpler OSs like CP/M (1974) and PC-DOS (1981).



History

4. Fourth Generation (1980– Present)

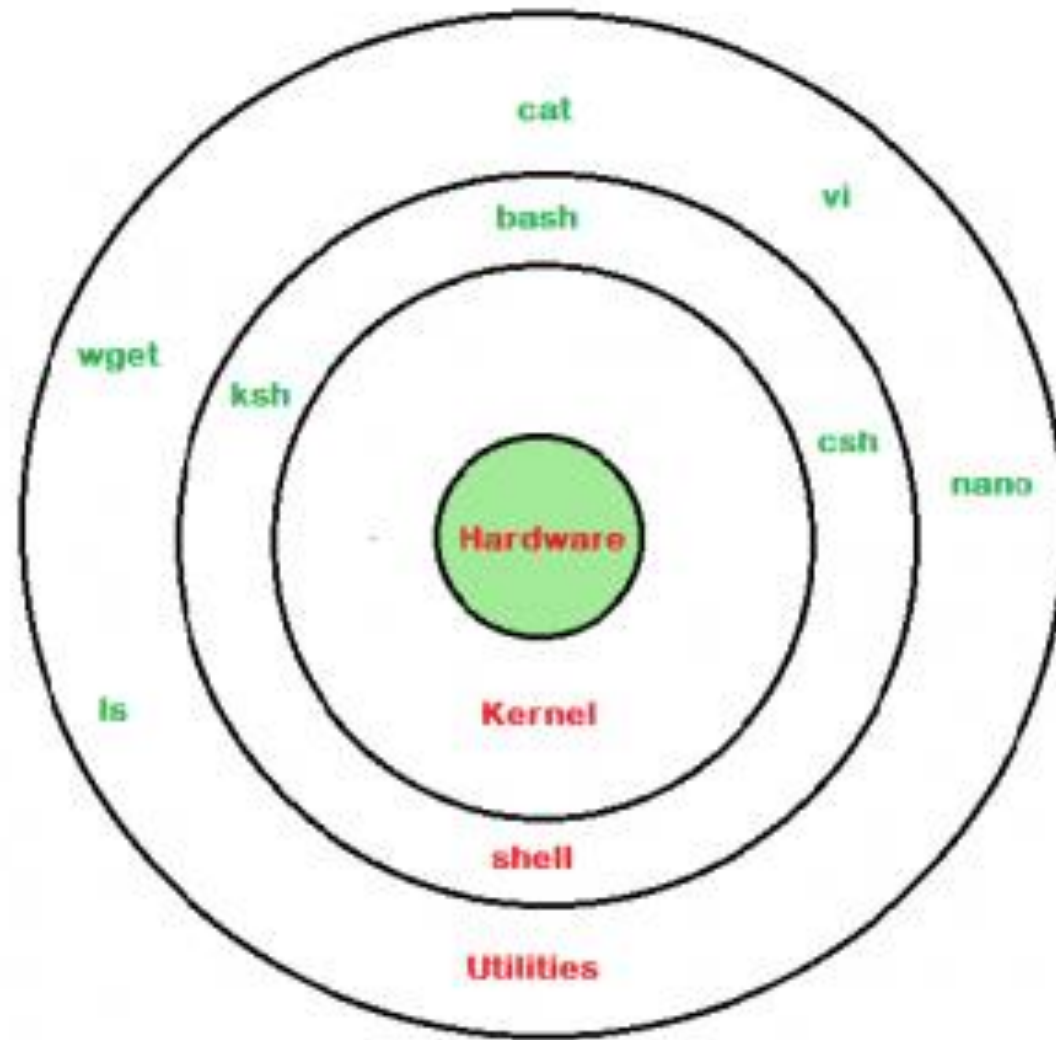
- GUIs gained popularity with systems like Apple Macintosh (1984) and Microsoft Windows (1985).
- Linux (1991) pioneered open-source development.
- Mobile OSs like iOS (2007) and Android (2008) have become dominant.



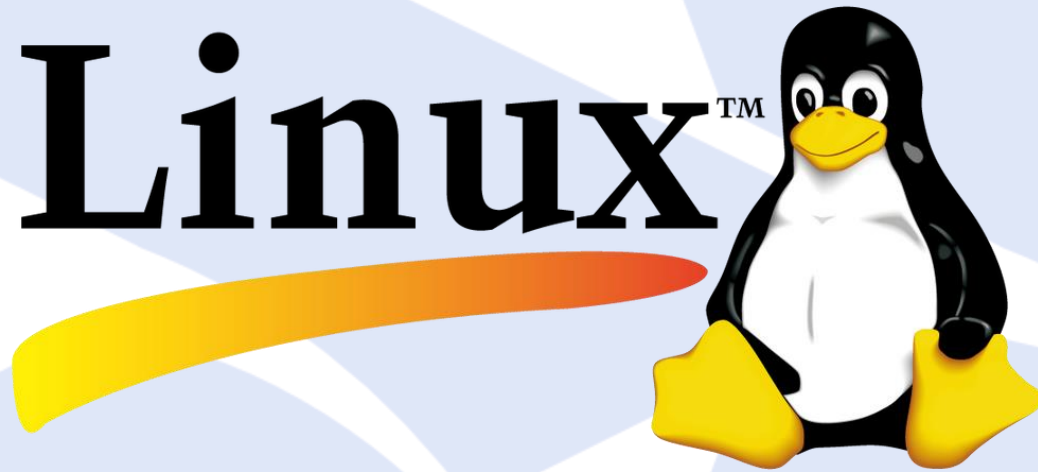


What IS Linux?

Linux Operating System Architecture



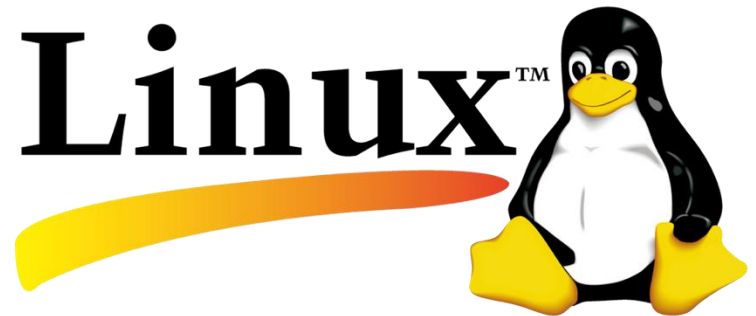
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Why Should You learn about Linux?

Linux

- 1 Open Source
- 2 Powerful CLI
- 3 Modular



You Are Already Using Linux



Open-Source Software

is software that is distributed with its source code.

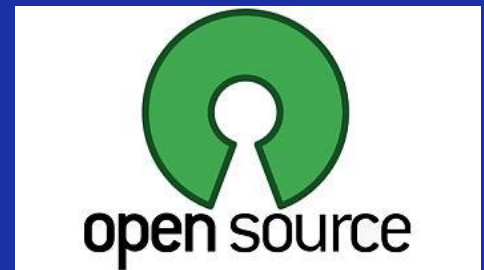
The Goal Not:

you use the application for free

The Goal Is:

Anyone can:

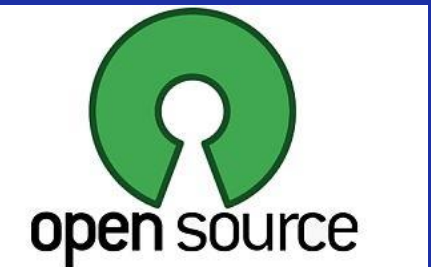
- Use
- Study
- Modify
- Share



Open-Source Software

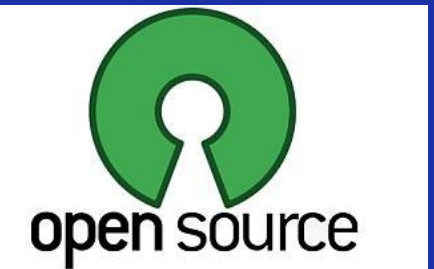
Benefits:

- Control
- Training
- Stability
- Security



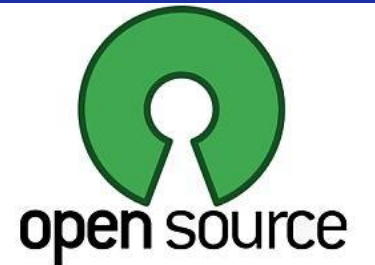
Types Of License

- **Permissive**
- **Copyleft**

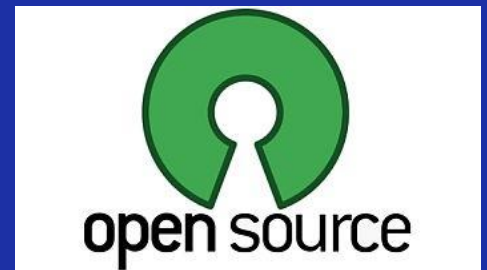


Licenses

- **MIT License**
- **GNU General Public License (GPL) 2.0**—this is more restrictive and requires that copies of modified code are made available for public use
- **Apache License 2.0**
- **GNU General Public License (GPL) 3.0**
- **BSD License 2.0 (3-clause, New or Revised)**—this is less restrictive

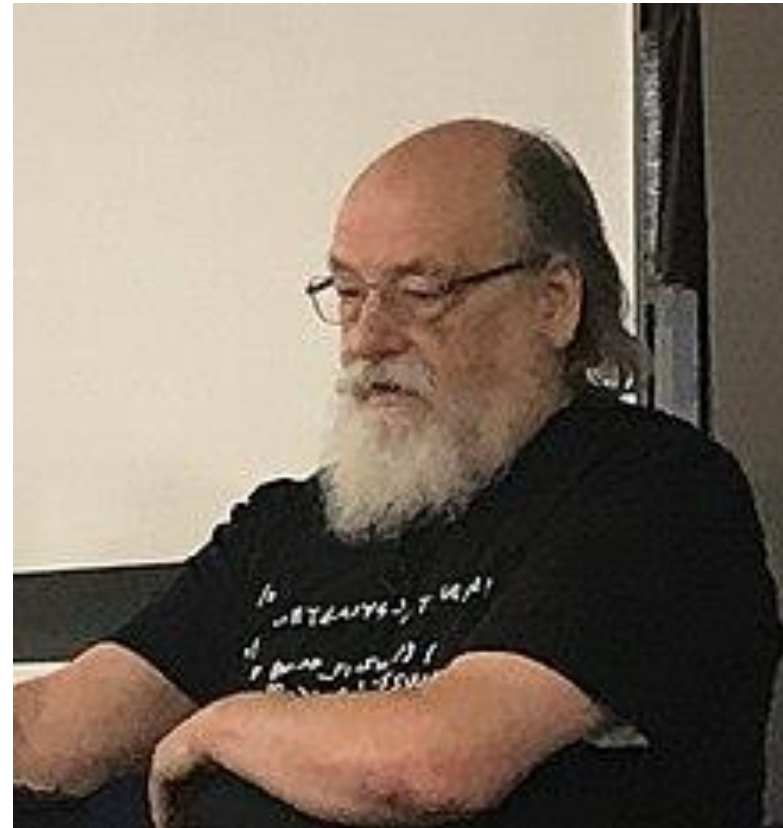


Who Develop Open-Source Software



Ken Thompson

- **Creator of Unix Operating System.**
- **Creator of B Language.**
- **UTF-8 Encoding (1992).**
- **Creator of Go Lang.**



Richard Stallman

- **Founder of the Free Software Movement.**
- **Creator of GNU Project in 1983**
- **Creator GNU General Public License (GPL)**
- **Initiator of Free Software Foundation (FSF)**
- **Developer of Emacs, GCC, and other GNU tools**



Linus Torvalds

- **Creator of the First Linux Kernel 1991**
- **Creator of Git, the distributed version control system**



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Distributions

Distribution

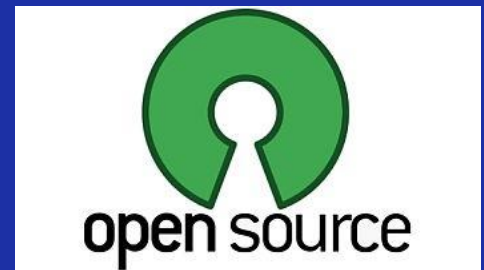
- 1 Linux Kernel
- 2 Supporting user Program and Libraries



Distribution

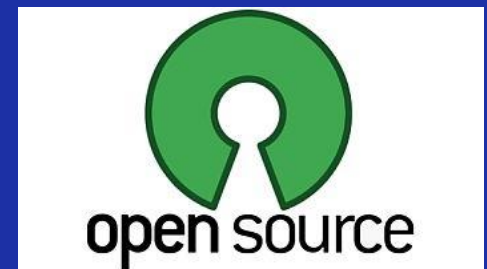
Common Characteristics :

- Linux Kernel and user space programs.
- Either Single purpose or include 1000s of programs.
- Must provide means of installing and updating the distribution and its components.
- Maintainers provide support for distro and open-source Community.



Distribution families.

three main “families” of Linux distributions: Debian, Red Hat, and SUSE.





takethenotes.com

Types of Linux Distributions

Desktop Linux



Server Linux



Red hat



debian



CentOS

Embedded Linux

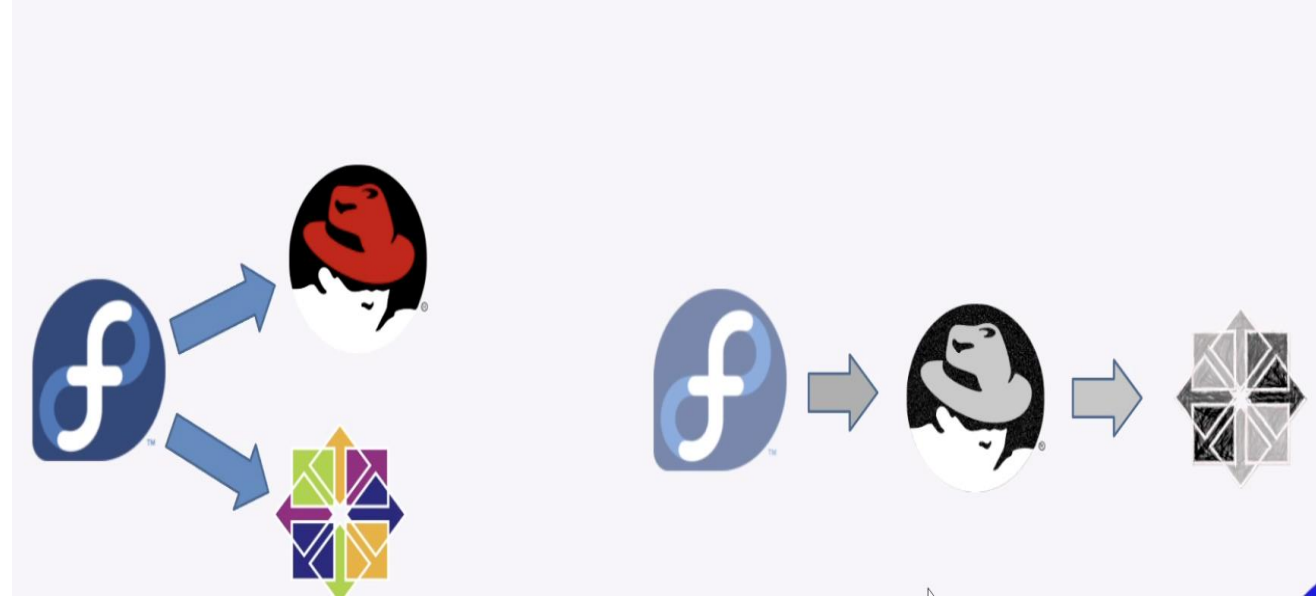


Raspberry Pi OS

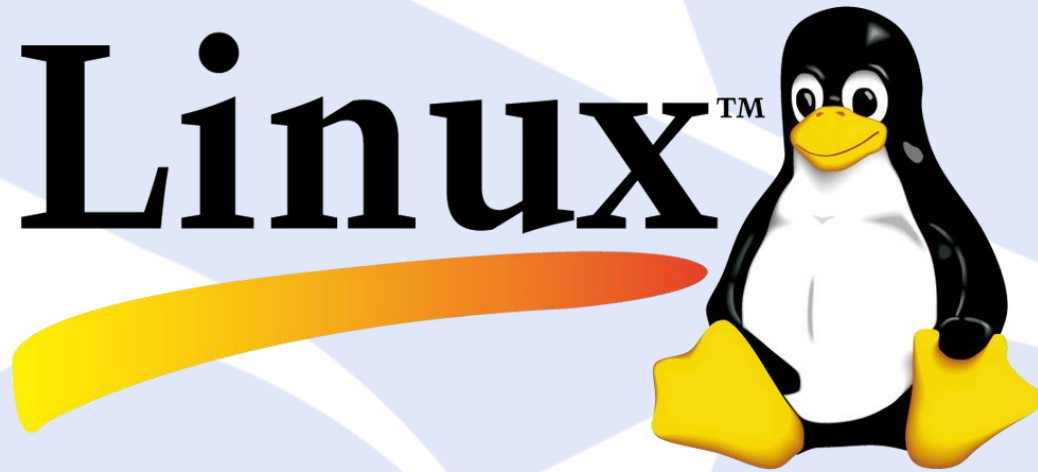
yocto
PROJECT

takethenotes.com

Upstream & Stream



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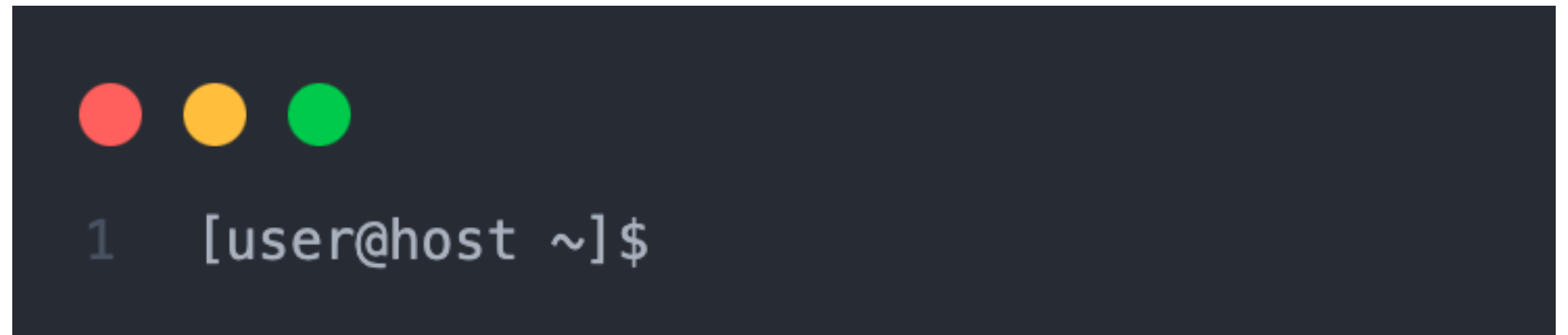


Accessing The CLI

Shell

In **Red Hat Enterprise Linux (RHEL)**,
the default shell for users is the **GNU Bourne-Again Shell (Bash)**.

Regular User:

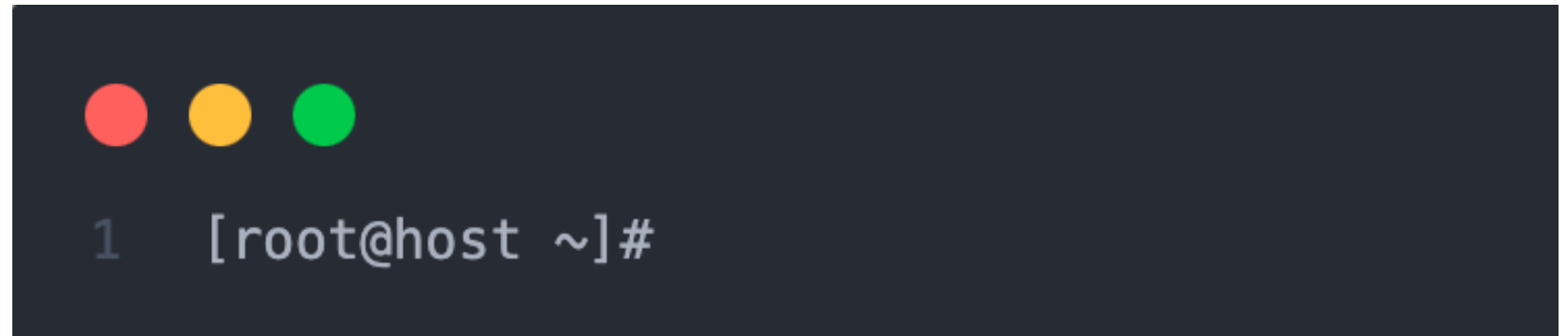
A dark-themed terminal window with three colored window control buttons (red, yellow, green) in the top left corner. The prompt shows a regular user at a host in their home directory.

```
1 [user@host ~]$
```

Shell

In **Red Hat Enterprise Linux (RHEL)**,
the default shell for users is the **GNU Bourne-Again Shell (Bash)**.

Root User (Administrator):

A terminal window with a dark background. At the top left, there are three colored circles: red, yellow, and green. Below them, the text "[root@host ~]#" is displayed in a light gray font, indicating the root user's prompt.

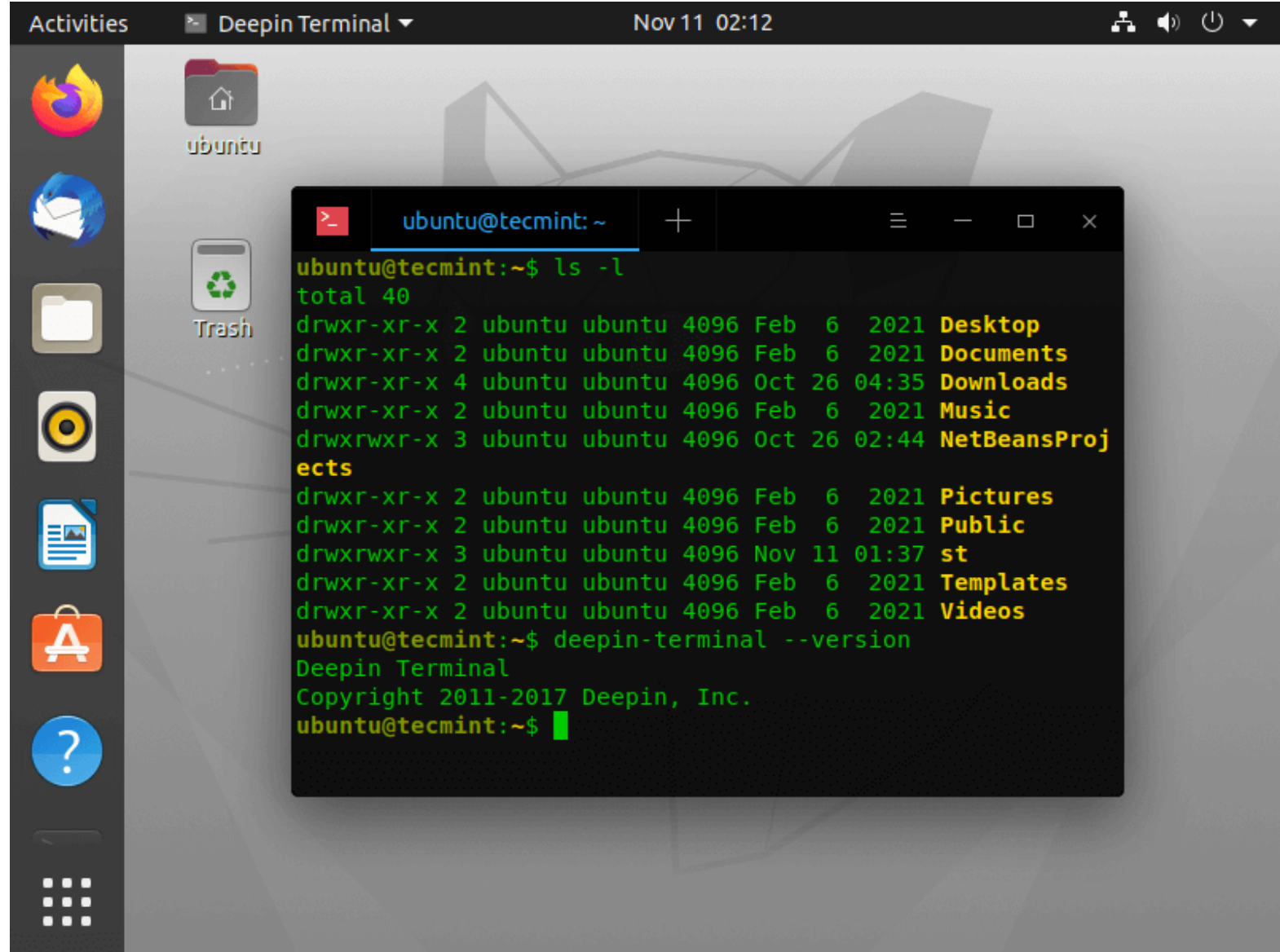
```
1 [root@host ~]#
```


Shell

A program that processes commands and returns output (e.g., Bash in Linux).

A program that runs a shell, allowing users to interact with it.

Terminal

A screenshot of a Linux desktop environment. The desktop background is a light gray with a faint geometric pattern. On the left side, there is a vertical dock with several application icons: Firefox, a mail client, a file manager, a music player, a document viewer, and an app store. The top of the screen features a panel with the 'Activities' button, a window title bar for 'Deepin Terminal', the date and time 'Nov 11 02:12', and system status icons for network, volume, and power. A 'Deepin Terminal' window is open in the foreground, displaying a shell prompt 'ubuntu@tecmint: ~'. The terminal shows the output of the 'ls -l' command, listing the contents of the home directory with permissions, owner, group, size, date, and name. The output includes Desktop, Documents, Downloads, Music, NetBeansProjects, Pictures, Public, st, Templates, and Videos. Below this, the terminal shows the command 'deepin-terminal --version' and its output, which includes the version and copyright information for Deepin Terminal.

```
ubuntu@tecmint: ~$ ls -l
total 40
drwxr-xr-x 2 ubuntu ubuntu 4096 Feb  6  2021 Desktop
drwxr-xr-x 2 ubuntu ubuntu 4096 Feb  6  2021 Documents
drwxr-xr-x 4 ubuntu ubuntu 4096 Oct 26 04:35 Downloads
drwxr-xr-x 2 ubuntu ubuntu 4096 Feb  6  2021 Music
drwxrwxr-x 3 ubuntu ubuntu 4096 Oct 26 02:44 NetBeansProj
ects
drwxr-xr-x 2 ubuntu ubuntu 4096 Feb  6  2021 Pictures
drwxr-xr-x 2 ubuntu ubuntu 4096 Feb  6  2021 Public
drwxrwxr-x 3 ubuntu ubuntu 4096 Nov 11 01:37 st
drwxr-xr-x 2 ubuntu ubuntu 4096 Feb  6  2021 Templates
drwxr-xr-x 2 ubuntu ubuntu 4096 Feb  6  2021 Videos
ubuntu@tecmint:~$ deepin-terminal --version
Deepin Terminal
Copyright 2011-2017 Deepin, Inc.
ubuntu@tecmint:~$
```

Basic Commands

whoami

1



```
1 ubuntu@ubuntu:~$ whoami
2 ubuntu
```

2



```
1 root@ubuntu:~# whoami
2 root
```

12

Basic Commands

hostnamectl

```
1 root@ubuntu:~# hostnamectl
2   Static hostname: ubuntu
3           Icon name: computer-vm
4           Chassis: vm
5           Machine ID: fef9c27f1bb74f5c81d1e799f5941e9c
6           Boot ID: aa88d6dab3cc4772a2f6a109a1cba3cd
7   Virtualization: qemu
8   Operating System: Ubuntu 22.04.2 LTS
9           Kernel: Linux 5.15.0-76-generic
10          Architecture: arm64
```

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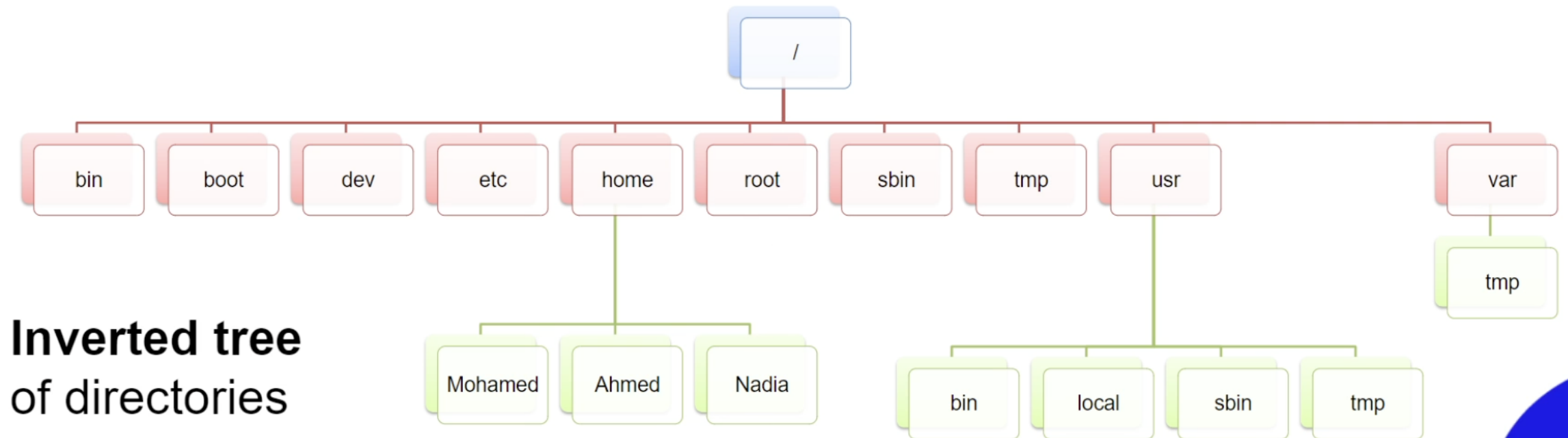


Managing files

copy, move, create, delete, organize files while working with bash

File system Hierarchy

THE FILE-SYSTEM HIERARCHY



Inverted tree
of directories

Root Directory /

Root Directory

Subdirectories of / are used for standardized purposes to organize files by type and purposes. This make it easier to find files.



/USR

User system Resources

Installed software, shared libraries, include files, and read-only program data.

User system Resources:

- /usr/bin
- /usr/sbin
- /usr/local
- /usr/lib
- /usr/share

/etc

**Configuration files specific to this system.
Stores system-wide configuration files.**

/etc/passwd

/etc/fstab

/etc/hosts



/var

Variable data specific to this system that should persist between boots.

Files that dynamically change, such as cache directories, Log files, printer-spoiled documents and website content may be found under /var

/var/log/syslog

/var/mail/

/var/www

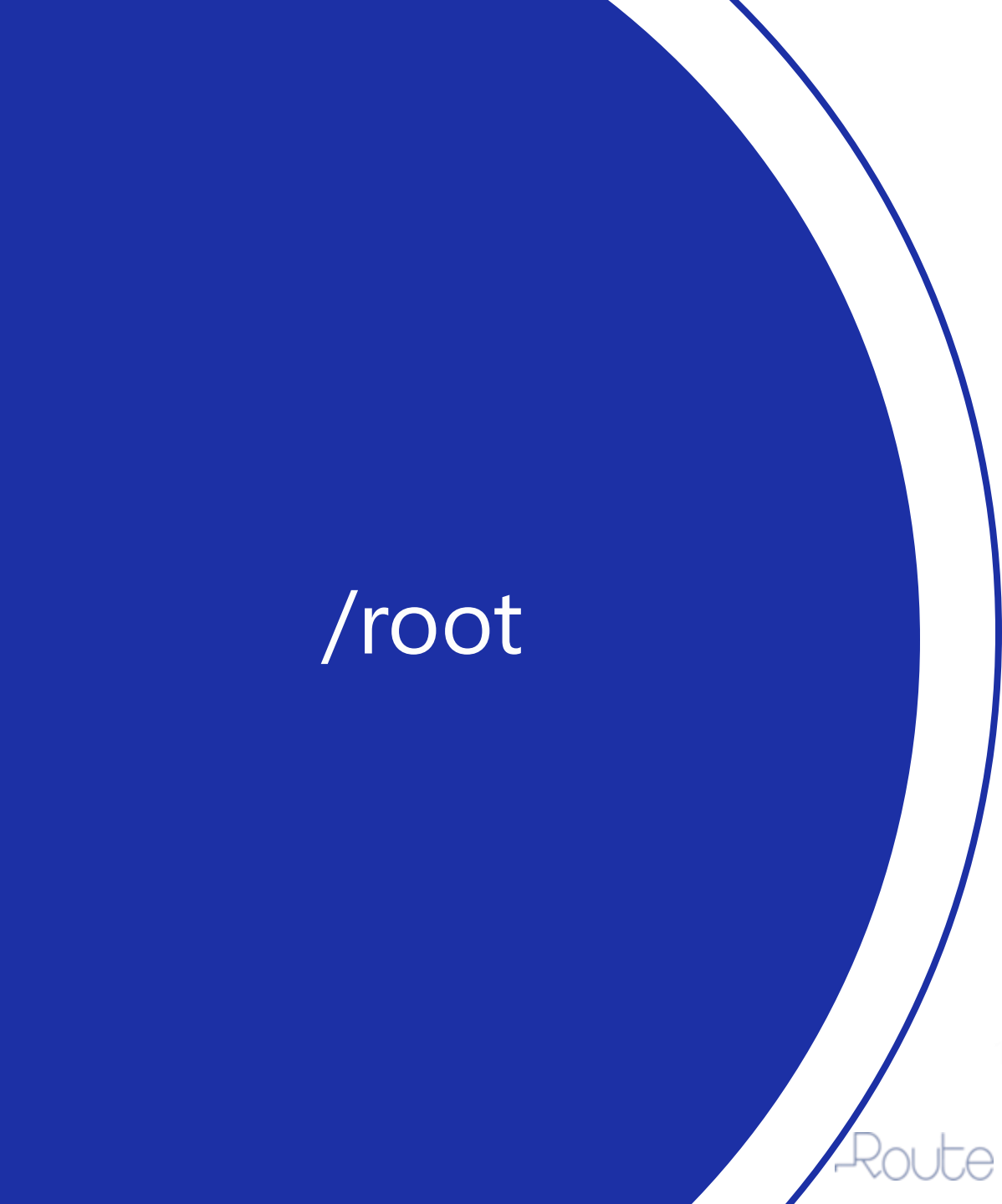


/run

Runtime data for processes started since the last boot. This includes process ID files and lock files. The content of this directory are recreated with reboot.

/Home

Home directories are the regular user store their personal data and configuration files.



/root

Home directory for the administrative superuser, root .

/tmp

A world-writable space for temporary files. Files which have not been accessed or modified for 10 days are deleted from the directory automatically.

Another Temporary Directory `/var/tmp` Files which have not been accessed or modified for 30 days are deleted from the directory automatically.

/boot

Files needed in order to start the boot process.

including the Linux kernel (vmlinuz).

Common files:

initrd.img – Temporary root filesystem during boot.

grub/ – GRUB bootloader files.



/dev

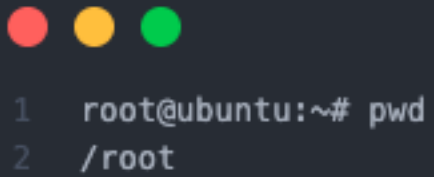
Special device files that are used by system to access the hardware.

/dev/sda # Primary disk drive

/dev/null # Null device (discard output)

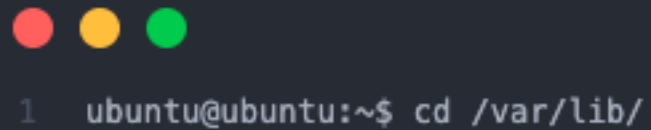
Directory

PWD = Print Working Directory



```
1 root@ubuntu:~# pwd
2 /root
```

cd = Change Directory



```
1 ubuntu@ubuntu:~$ cd /var/lib/
```

Relative Path VS Absolute Path

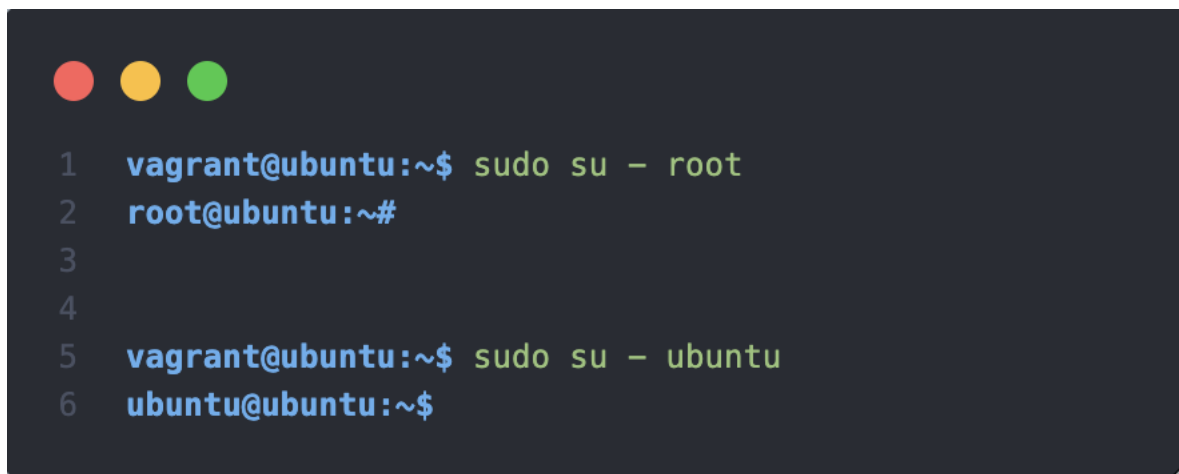
An absolute path is the location of a file or directory specified from the root directory. It represents the complete path from the start of the filesystem from the root directory.

Relative Path VS Absolute Path

A relative path is the location of a file or directory relative to your current working directory. Unlike an absolute path, the relative path doesn't start with a slash or drive letter since it doesn't start from the root directory.

Changing the user

su



```
1 vagrant@ubuntu:~$ sudo su - root
2 root@ubuntu:~#
3
4
5 vagrant@ubuntu:~$ sudo su - ubuntu
6 ubuntu@ubuntu:~$
```

Commands to Manage Files

Navigate Files

Ls , cd , pwd

Manage

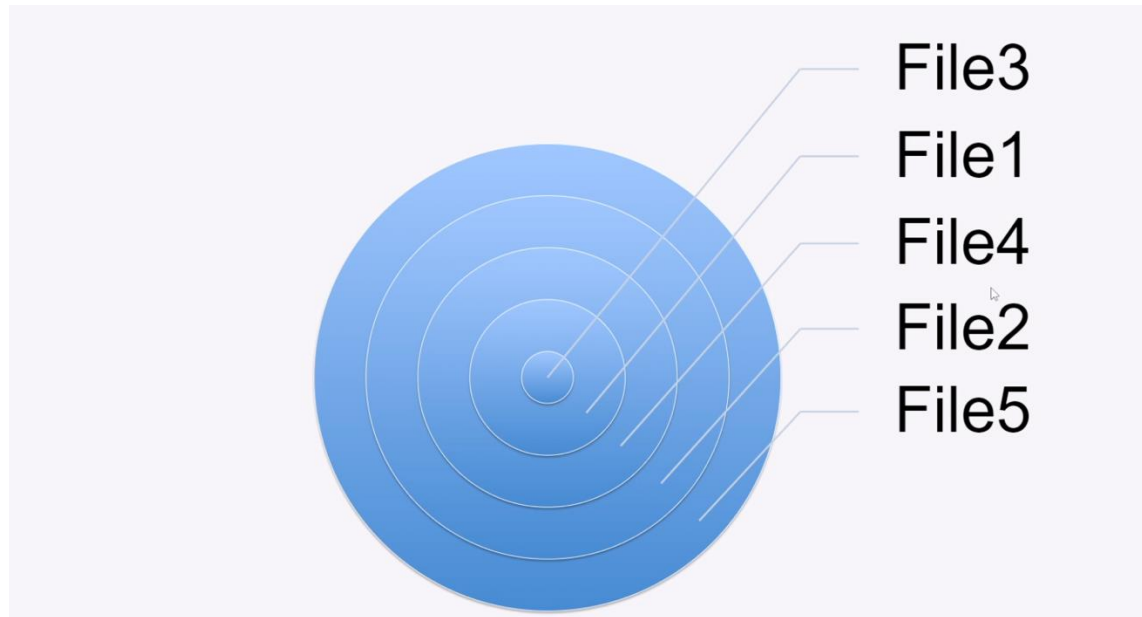
Mkdir, cp, mv, rm, touch, hard links, soft links

Edit & view

Cat, vim, vi, nano

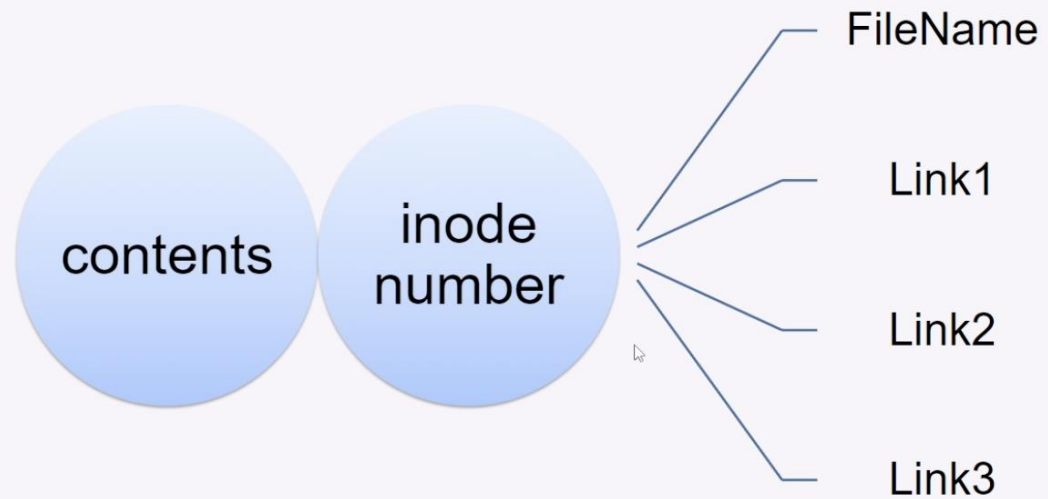
Links Between Files (Soft, hard)

File in disk



Hard Links

Hard Links



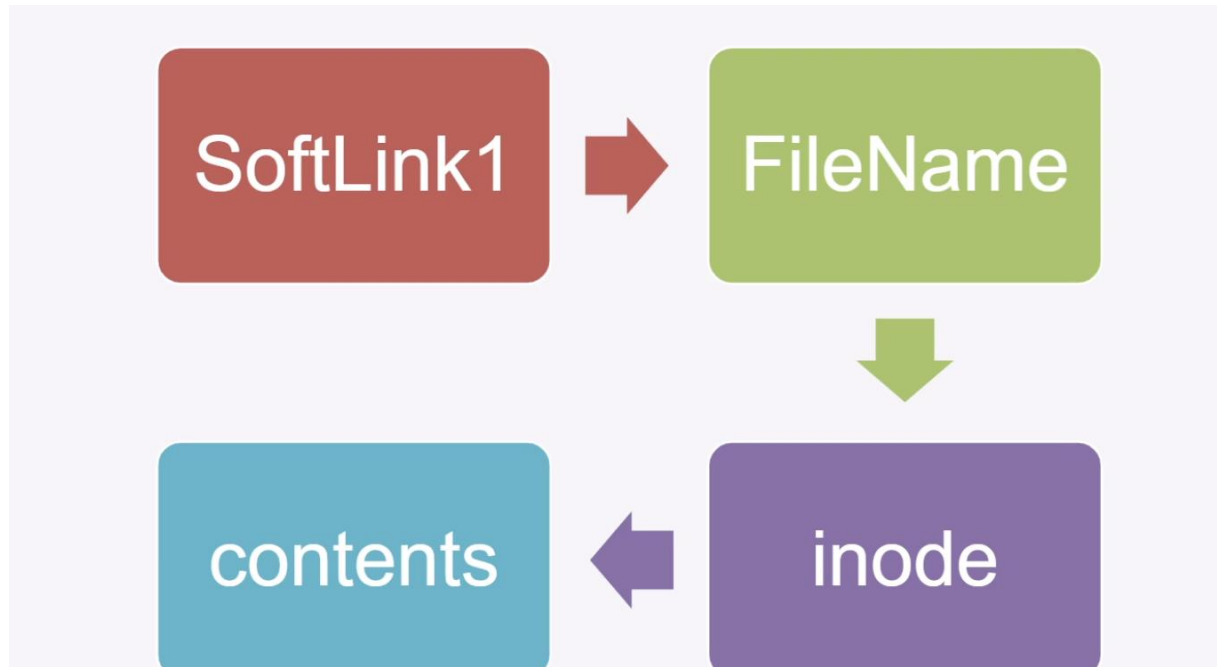
Hard Links

**Only on Files
in the same Filesystem
same inode**



```
1 ln original.txt hard_link.txt
2 rm original.txt
```

Soft Links



Linking Command

```
[user@host]$ ln FileName HardLink  
[user@host]$ ln -s FileName SoftLink  
[user@host]$ ls -l .  
[user@host]$ ls -il  
[user@host]$ df
```

Expansion



```
1  Touch USER-{A..C}-{1..3}
2  ls USER-B-*
3  ls USER-[AC]-*
4  Touch USER-{0..9}{0..9}{0..9}
5  ls *3*
6  ls 9*8
7  ls users-??[37]
8  RESIGNED=USER-800
9  echo $RESIGNED
10 echo date vs echo $(date)
```

Manual Pages

MAN Command

Grep

Pipe "|"

grep

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Lab 1

Lab 1



1 Lab 1: Linux Basics

2

- 3 o Navigate the filesystem using ls, cd, and pwd.
- 4 o Create, move, copy, and delete files and directories using mkdir, cp, mv, and rm.
- 5 o View and edit files with cat, nano, and vi.

Lab 1 – Solution

